

Question 1 | Correct Mark 1.00 out of 1.00 [Flag question](#)

Convert the following algorithm into a program and find its time complexity using the counter method.

void function (int n)

```
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     int i= 1;
6     int s= 1;
7     int count = 3;
8     scanf("%d", &n);
9     while(s <= n)
10 {
11     i++;
12     s += i;
13     count += 3;
14 }
15 printf("%d", count);
16 return 0;
17 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     int count=0;
5     count++;
6     scanf("%d",&n);
7     if(n==1)
8     {
9         count++;
10        count++;
11    }
12    else
13    {
14        for(int i=1; i<=n; i++)
15        {
16            count+=2;
17            for(int j=1; j<=n; j++)
18            {
19                count+=1;
20                count++;
21                count++;
22                break;
23            }
24        }
25    }
26    count++;
27    printf("%d",count);
28    return 0;
29 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

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Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>  
2 int main(){  
3     int n;  
4     int count=0;  
5     scanf("%d",&n);  
6     for (int i = 1; i <= n; ++i){  
7         count++;  
8         if (n % i == 0){  
9             count++;  
10            count++;  
11        }  
12        else{  
13            count++;  
14        }  
15    }  
16    count++;  
17    printf("%d", count);  
18    return 0;  
19 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable
Answer:

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     int count=0;
5     scanf("%d",&n);
6     int c= 0;
7     count++;
8     for(int i=n/2; i<n; i++){
9         count++;
10        for(int j=1; j<n; j = 2 * j){
11            count++;
12            for(int k=1; k<n; k = k * 2){
13                count++;
14                c++;
15                count++;
16            }
17            count++;
18        }
19        count++;
20    }
21    count++;
22    printf("%d",count);
23    return 0;
24 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

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Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     int count=0;
5     scanf("%d",&n);
6     int rev = 0, remainder;
7     count++;
8     while (n != 0){
9         count++;
10        remainder = n % 10;
11        count++;
12        rev = rev * 10 + remainder;
13        count++;
14        n /= 10;
15        count++;
16    }
17    count++;
18    count++;
19    printf("%d",count);
20    return 0;
21 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.